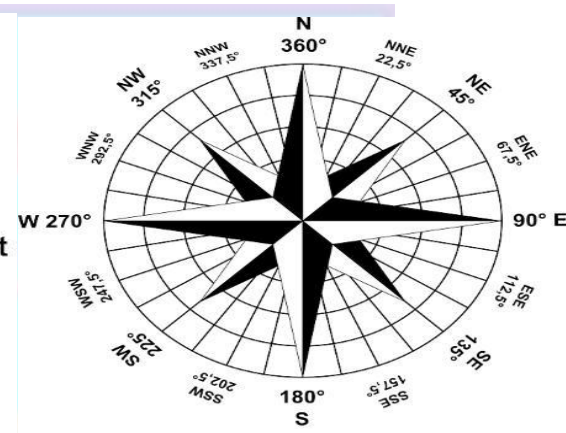
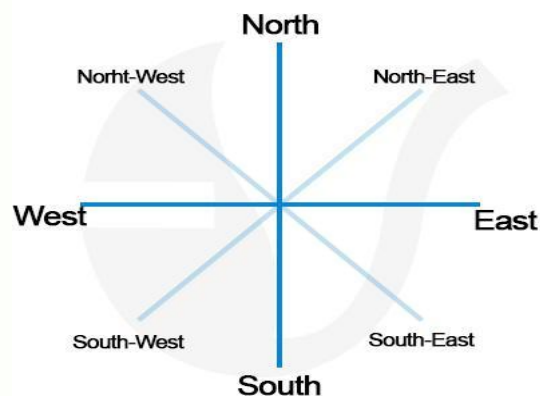
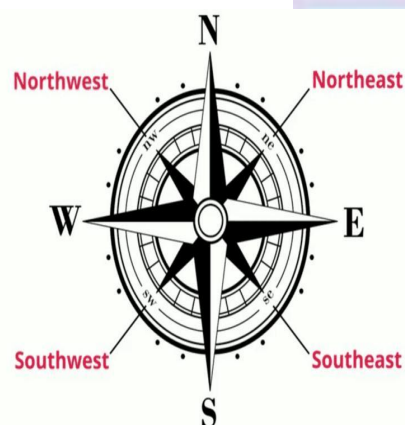


WEEK 4: BEARINGS AND DISTANCES I

TOPIC: Cardinal Points; Bearings; Scale Drawing; Distances and Navigation

A. CARDINAL POINTS (BASIC DIRECTION SYSTEM)



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Definition:

Cardinal points are the **four main directions** used in navigation and mapping.

The Four Main Directions:

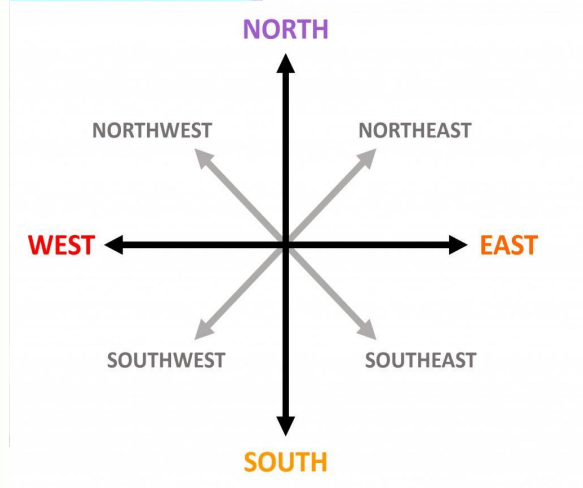
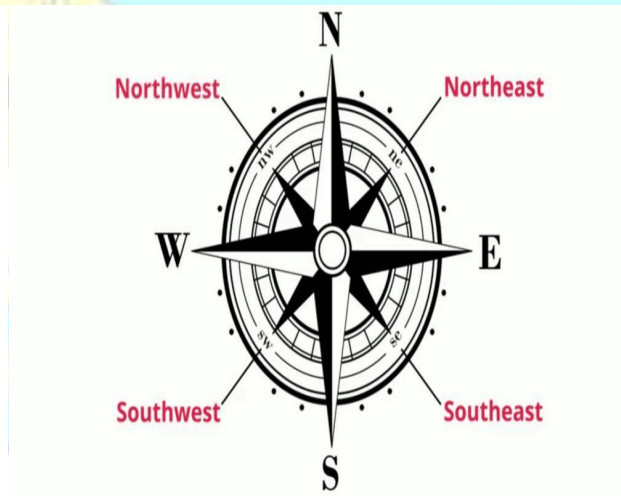
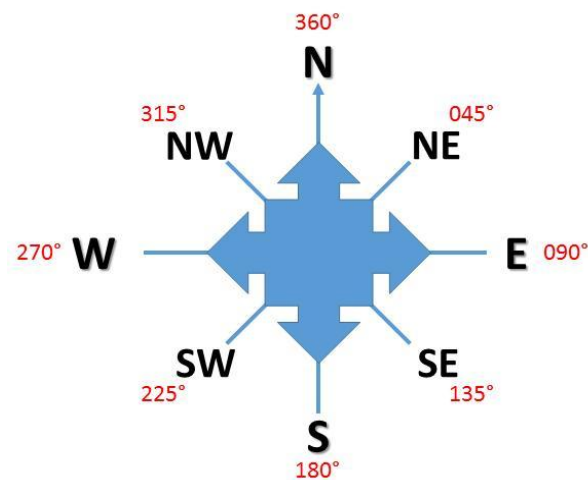
- North (N) = 0° / 360°
- East (E) = 90°

- South (S) = 180°
- West (W) = 270°

Key Rule:

Bearings are ALWAYS measured **clockwise from North**

B. INTERCARDINAL POINTS (SECONDARY DIRECTIONS)



The 8 Main Directions:

Direction Bearing

North 000°



Direction Bearing

Northeast 045°

East 090°

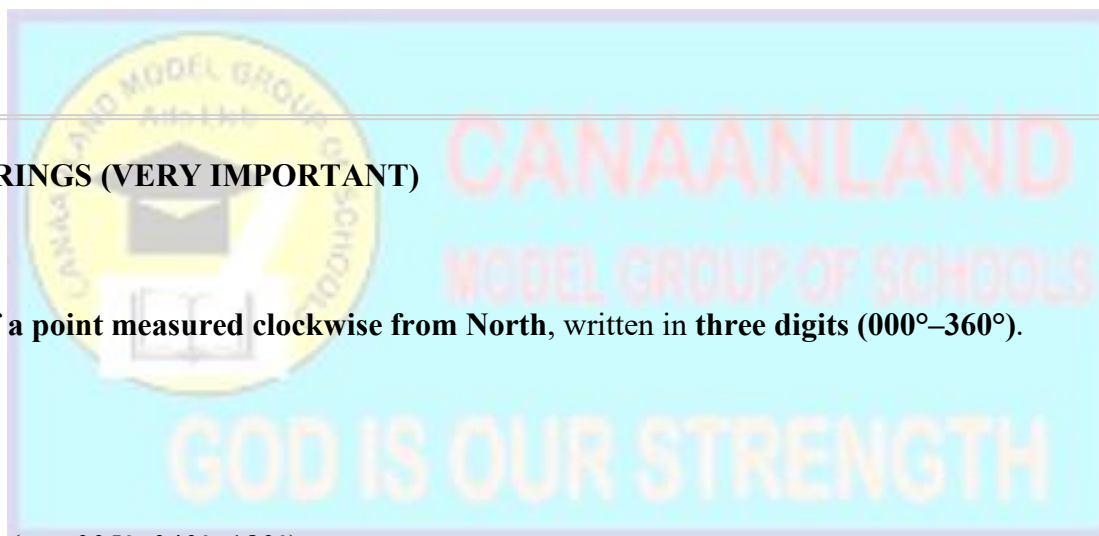
Southeast 135°

South 180°

Southwest 225°

West 270°

Northwest 315°



C. THREE-FIGURE BEARINGS (VERY IMPORTANT)

Definition:

A bearing is the **direction of a point measured clockwise from North**, written in **three digits (000°–360°)**.

Rules:

1. Start from North (0°)
2. Move clockwise
3. Always write 3 digits (e.g. 005°, 040°, 180°)

EXAMPLE 1: BASIC DIRECTIONS TO BEARINGS

Convert to bearings:

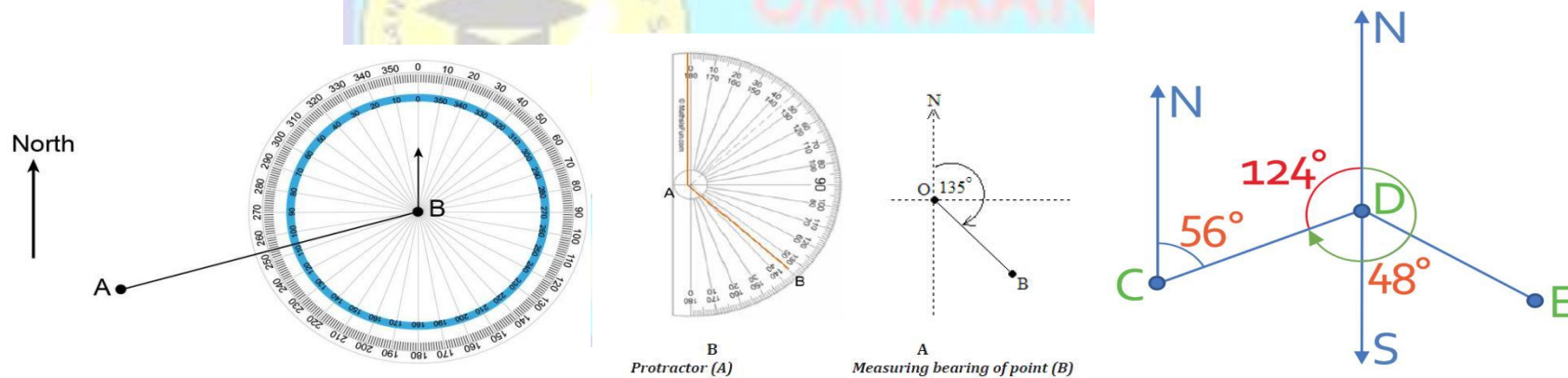
- North = 000°
- East = 090°

- South = 180°
- West = 270°

EXAMPLE 2: INTERCARDINAL POINTS

- NE = 045°
- SE = 135°
- SW = 225°
- NW = 315°

D. MEASURING BEARINGS (STEP-BY-STEP)



EXAMPLE 3:

“ 30° East of North”

Solution:



- Start at North = 0°
- Move 30° clockwise toward East

Bearing = **030°**

EXAMPLE 4:

“ 40° South of East”

Solution:

- East = 90°
- Move 40° toward South (clockwise)

$$90^\circ + 40^\circ = \mathbf{130^\circ}$$

EXAMPLE 5:

“ 25° West of South”

Solution:

- South = 180°
- Move 25° toward West

$$180^\circ + 25^\circ = \mathbf{205^\circ}$$

EXAMPLE 6 (TRICKY):





“50° North of West”

- West = 270°
- Move 50° toward North (anticlockwise)

$$270^\circ + 50^\circ = 320^\circ \text{ (correct clockwise equivalent)}$$

E. BACK BEARINGS

RULE:

- If bearing $< 180^\circ \rightarrow$ add 180°
- If bearing $> 180^\circ \rightarrow$ subtract 180°

EXAMPLE 7:

Bearing A to B = 060°

$$\text{Back bearing} = 060^\circ + 180^\circ = \mathbf{240^\circ}$$

EXAMPLE 8:

Bearing P to Q = 215°

$$\text{Back bearing} = 215^\circ - 180^\circ = \mathbf{035^\circ}$$

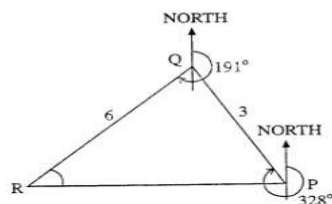
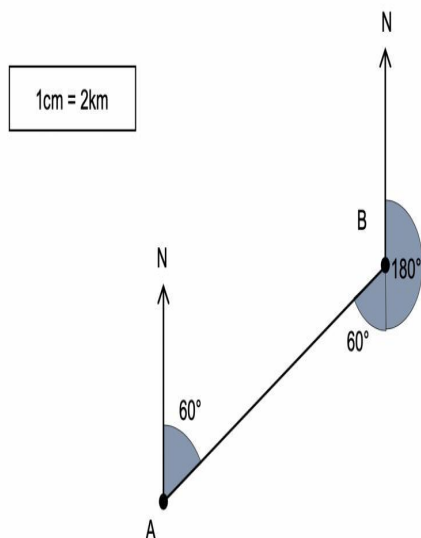
EXAMPLE 9:

Bearing X to Y = 130°



Back bearing = $130^\circ + 180^\circ = 310^\circ$

F. LOCATING POSITIONS (BEARING + DISTANCE)

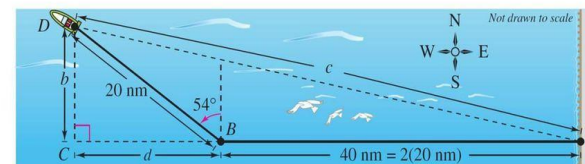


The diagram shows three villages P, Q and R. The bearing of Q from P is 328° and the bearing of R from Q is 191° . $PQ = 3$ km and $QR = 6$ km.

- (a) Calculate
- $\angle PQR$,
 - the distance Q is north of P,
 - PR .
- (b) Find the bearing of R from P, giving your answer correct to the nearest degree.

Example 5 – Finding Directions in Terms of Bearings

A ship leaves port at noon and heads due west at 20 knots, or 20 nautical miles (nm) per hour. At 2 P.M. the ship changes course to $N 54^\circ W$, as shown below. Find the ship's bearing and distance from the port of departure at 3 P.M.



METHOD:

To locate a point, you need:

- Starting point
- Bearing
- Distance
- Scale



EXAMPLE 10:

From A:

- B is 50 km on bearing 070°

SCALE: 1 cm = 10 km

Step-by-step:

1. Draw point A
2. Draw North line
3. Measure 70° clockwise
4. Convert distance: $50 \div 10 = 5$ cm
5. Mark 5 cm along direction

B is located

EXAMPLE 11:

From town T:

- V is 30 km on bearing 215°

Scale: 1 cm = 5 km

$$30 \div 5 = 6 \text{ cm}$$

Draw 6 cm at 215° from T





G. FINDING DISTANCES USING SCALE / PYTHAGORAS

EXAMPLE 12:

- A is 40 km North of B
- C is 30 km East of B

Find AC:

Step 1: Form triangle

Right-angled triangle formed

Step 2: Apply Pythagoras

$$AC^2 = 40^2 + 30^2$$

$$AC^2 = 1600 + 900$$

$$AC^2 = 2500$$

$$AC = \sqrt{2500} = 50 \text{ km}$$



EXAMPLE 13 (SCALE METHOD):

If AC = 5 cm on drawing and scale = 10 km/cm:

$$AC = 5 \times 10 = 50 \text{ km}$$

H. FINDING BEARINGS USING SCALE DRAWING



EXAMPLE 14:

- A is 50 km North of B
- C is 40 km East of B

Find bearing of C from A

Steps:

1. Draw accurate diagram
2. Join A to C
3. Measure clockwise from North at A

Bearing $\approx 129^\circ$

EXAMPLE 15:

- A = 25 km on 040°
- B = 35 km on 150°

Find bearing of B from A

Construct diagram
Measure angle

Result $\approx 168^\circ$

I. COMBINED NAVIGATION PROBLEMS





EXAMPLE 16 (2-STEP JOURNEY):

- 8 km on 060° to P
- 6 km on 150° to Q

Find:

- Distance CQ
- Bearing of Q from C

Method:

1. Draw both paths using scale
2. Join C to Q
3. Measure

CQ $\approx$ 12 km

Bearing $\approx 094^\circ$



EXAMPLE 17 (RETURN JOURNEY):

Bearing $A \rightarrow B = 135^\circ$

Return bearing:

$$135^\circ + 180^\circ = \mathbf{315^\circ}$$



J. COMMON EXAM MISTAKES

- ✗ Measuring anticlockwise
- ✗ Forgetting North reference line
- ✗ Not using 3-digit format
- ✗ Wrong scale conversion
- ✗ Confusing elevation with bearing

K. PRACTICAL APPLICATIONS

- Aircraft navigation
- Ship routing
- GPS mapping
- Military operations
- Land surveying



L. CLASSWORK

1. Define bearing
2. State 4 cardinal points
3. Convert NE to bearing
4. What is back bearing of 075° ?
5. What is due South bearing?



6. Why do we use scale drawings?
7. Find distance using Pythagoras (40 km, 30 km)
8. Convert $S30^{\circ}E$ to bearing
9. What instrument measures bearings?
10. Draw a diagram showing 125° bearing

